

RAY VERMA

rayverma@nyu.edu
linkedin.com/in/ray-verma
ray-verma.github.io

EDUCATION

New York University

Aug. 2023 – Jul. 2027

B.S., double major in Computer Science and Mathematics

Abu Dhabi, UAE

GPA: 3.95/4.00. Awarded scholarship of \$280,000, covering full tuition expense; Global Program in Abu Dhabi, New York, Paris, and Los Angeles.

RESEARCH EXPERIENCE

Center for AI and Robotics, NYU | *Research Assistant*

Feb. 2026 – Present

Abu Dhabi, UAE

- Decentralized morphogenesis: Developing local controllers for homogeneous robot swarms that form, reconfigure and recover global shapes using only neighborhood observations.
- Dynamic target assignment: Designing GoalNCA, a neural cellular automata approach for decentralized target discovery and scalable lattice formation control.
- Learned Morphogenesis: Developing NavCA, an end-to-end equivariant graph cellular automata controller that generalizes across target shapes without global coordinates.

AGI Lab, Stanford University | *Research Assistant*

Aug. 2025 – Present

Remote / California, USA

- Geometry-aware representations: Developed Random Voronoi Networks, interpretable feature maps that induce geometric bias in simple models and improve F1 performance by up to 40%.
- Unsupervised dataset embeddings: Developed Expand and Reduce, a schema-agnostic collection-level representation for AutoML that outperformed supervised benchmarks by 18%.
- Tabular foundation models: Investigating Prior-Fitted Networks for data-efficient prediction and transfer from synthetic data to heterogeneous real-world datasets.

Genetic Heritage Group, NYU | *Research Assistant*

Aug. 2024 – Sep. 2026

Abu Dhabi, UAE

- Population-scale microbiome analysis: Built an archetypal-analysis and visualization toolkit with a 20× faster dimensionality-reduction workflow, processing more than 30,000 samples.
- Causal interaction discovery: Adapted HaLLA and PCMCI+ for time-series bacteria-virus analysis, identifying ecological dependencies relevant to drug discovery for antimicrobial-resistant pathogens.
- Cross-dataset validation: Collaborated with UC Santa Cruz across four microbiome datasets to evaluate community structure and cumulative biological effects under a common workflow.

INDUSTRY EXPERIENCE

Galatea Bio | *AI Engineer Intern*

Jun. 2025 – Aug. 2025

Remote / Florida, USA

- Memory-aware scheduling: Engineered RAM-efficient execution for chromosome-level genomic workflows using symbolic regression to predict task demand and applied conformal bounds to limit over-scheduling.
- Adaptive task packing: Developed a resource-aware orchestration methods that reduced multi-task runtime by 60% while increasing system utilization by 75%.
- Workflow reliability: Validated scheduling policies through simulation and visualization, refactored core components and hardened execution with Prefect and GitHub Actions.

Emerton Data | *AI Engineer Intern*

Jan. 2025 – May 2025

Paris, France

- Low-resource reasoning: Fine-tuned Qwen and Llama models for chain-of-thought translation with fewer than 10,000 samples using GRPO reinforcement learning.
- Multi-reward optimization: Designed a dynamically weighted objective over seven reward functions and a reproducible 10-day remote-GPU training pipeline for a Government of Monaco project.
- Reproducible experimentation: Instrumented training with Hugging Face, Unsloth and Weights & Biases to compare reward schedules, checkpoints and stability.

PAPERS & PUBLICATIONS

GoalNCA: Neural Cellular Automata for Distributed Target Discovery in Swarm Formation Control *ongoing*

Ray Verma, Eliseo Ferrante, Anthony Tsez.

NavCA: Graph Cellular Automata in Robot Control for Swarm Morphogenesis *ongoing*

Ray Verma, Eliseo Ferrante, Anthony Tsez.

Archetypal Analysis for Microbiome Research *ongoing*

Ray Verma, Aashish Jha, Anique Ahmad, Alexander G. Ioannidis.

Random Voronoi Networks *preprint*

Ray Verma, Pau Mateo Bernadó, Daniel Mas Montserrat, David Bonet, Marçal Comajoan Cara, Alexander G. Ioannidis.

Expand and Reduce: Dimensionality-Agnostic Dataset Embeddings *preprint*

Ray Verma, Míriam Barrabés, David Bonet, Marçal Comajoan Cara, Daniel Mas Montserrat, Alexander G. Ioannidis.

Efficient Chromosome Parallelization for Precision Medicine Genomic Workflows *AAAI 2026*

Daniel Montserrat, **Ray Verma**, Míriam Barrabés, Francisco M. de la Vega, Carlos D. Bustamante, Alexander G. Ioannidis.

🏆 Deployed Application Award

PRESENTATIONS

Capstone Research Proposal @ NYU Abu Dhabi April. 2026

Proposed “*Decentralized Morphogenesis with Neural Cellular Automata*,” outlining scalable swarm robotics architectures for coordinate-free shape formation and dynamic target assignment.

Paper Presentation @ AAAI 2026, Singapore Feb. 2026

Presented “*Efficient Chromosome Parallelization for Precision Medicine Genomic Workflows*” (Deployed Application Award winner) at the 40th AAAI Conference on Artificial Intelligence.

Invited Seminar @ NYU Jun. 2024

Delivered a technical deep-dive on state-of-the-art generalizable 3D Gaussian Splatting literature to Master’s research groups to facilitate integration into ongoing embodied AI pipelines.

SKILLS & INTERESTS

Programming Languages: Python, C++, MATLAB

Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, Weights & Biases

Research Methodologies: Neural & Graph Cellular Automata, Feature Encoding, Set Embedding, AutoML, Causal Inference, Symbolic Regression, Reinforcement Learning

Systems & Infrastructure: GitHub Actions, Prefect, Linux, HPC/Slurm Clusters, Distributed GPU Computing, Scientific Visualization

Languages: English (Fluent), Hindi (Fluent), French (Intermediate)

Interests: Origami, Drumming